Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim: To learn the Basics of Machine Learning**	
Experiment No: 01	**Date:**	**Enrollment No:92301733057**

Aim: To learn the Basics of Machine Learning

IDE: Google Colab

Theory:

Machine learning is a subset of artificial intelligence in the field of computer science that often uses statistical techniques to give computers the ability to "learn" (i.e., progressively improve performance on a specific task) with data, without being explicitly programmed. In the past decade, machine learning has given us self-driving cars, practical speech recognition, effective web search, and a vastly improved understanding of the human genome.

Machine learning tasks

Machine learning tasks are typically classified into two broad categories, depending on whether there is a learning "signal" or "feedback" available to a learning system:

- **Supervised learning:** The computer is presented with example inputs and their desired outputs, given by a "teacher", and the goal is to learn a general rule that maps inputs to outputs. As special cases, the input signal can be only partially available, or restricted to special feedback:
- **Semi-supervised learning:** the computer is given only an incomplete training signal: a training set with some (often many) of the target outputs missing.
- **Active learning:** the computer can only obtain training labels for a limited set of instances (based on a budget), and also has to optimize its choice of objects to acquire labels for. When used interactively, these can be presented to the user for labeling.
- **Reinforcement learning:** training data (in form of rewards and punishments) is given only as feedback to the program's actions in a dynamic environment, such as driving a vehicle or playing a game against an opponent.
- **Unsupervised learning:** No labels are given to the learning algorithm, leaving it on its own to find structure in its input. Unsupervised learning can be a goal in itself (discovering hidden patterns in data) or a means towards an end (feature learning).

Machine learning applications

In classification, inputs are divided into two or more classes, and the learner must produce a model that assigns unseen inputs to one or more (multi-label classification) of these classes. This is typically tackled in a supervised manner. Spam filtering is an example of classification, where the inputs are email (or other) messages and the classes are "spam" and "not spam". In regression, also a supervised problem, the outputs are continuous rather than discrete.

In clustering, a set of inputs is to be divided into groups. Unlike in classification, the groups are not known beforehand, making this typically an unsupervised task. Density estimation finds the distribution of inputs in

some space. Dimensionality reduction simplifies inputs by mapping them into a lower- dimensional space. Topic modeling is a related problem, where a program is given a list of human language documents and is tasked with finding out which documents cover similar topics.

Machine learning Approaches

Decision tree learning: Decision tree learning uses a decision tree as a predictive model, which maps observations about an item to conclusions about the item's target value. Association rule learning Association rule learning is a method for discovering interesting relations between variables in large databases.

Artificial neural networks

An artificial neural network (ANN) learning algorithm, usually called "neural network" (NN), is a learning algorithm that is vaguely inspired by biological neural networks. Computations are structured in terms of an interconnected group of artificial neurons, processing information using a connectionist approach to computation. Modern neural networks are non-linear statistical data modeling tools. They are usually used to model complex relationships between inputs and outputs, to find patterns in data, or to capture the statistical structure in an unknown joint probability distribution between observed variables.

Deep learning

Falling hardware prices and the development of GPUs for personal use in the last few years have contributed to the development of the concept of deep learning which consists of multiple hidden layers in an artificial neural network. This approach tries to model the way the human brain processes light and sound into vision and hearing. Some successful applications of deep learning are computer vision and speech recognition.

Inductive logic programming

Inductive logic programming (ILP) is an approach to rule learning using logic programming as a uniform representation for input examples, background knowledge, and hypotheses. Given an encoding of the known background knowledge and a set of examples represented as a logical database of facts, an ILP system will derive a hypothesized logic program that entails all positive and no negative examples. Inductive programming is a related field that considers any kind of programming languages for representing hypotheses (and not only logic programming), such as functional programs.

Support vector machines

Support vector machines (SVMs) are a set of related supervised learning methods used for classification and regression. Given a set of training examples, each marked as belonging to one of two categories, an SVM training algorithm builds a model that predicts whether a new example falls into one category or the other.

Clustering

Cluster analysis is the assignment of a set of observations into subsets (called clusters) so that observations within the same cluster are similar according to some pre designated criterion or criteria, while observations drawn from different clusters are dissimilar. Different clustering techniques make different assumptions on the

structure of the data, often defined by some similarity metric and evaluated for example by internal compactness (similarity between members of the same cluster) and separation between different clusters. Other methods are based on estimated density and graph connectivity. Clustering is a method of unsupervised learning, and a common technique for statistical data analysis.

Bayesian networks

A Bayesian network, belief network or directed acyclic graphical model is a probabilistic graphical model that represents a set of random variables and their conditional independencies via a directed acyclic graph (DAG). For example, a Bayesian network could represent the probabilistic relationships between diseases and symptoms. Given symptoms, the network can be used to compute the probabilities of the presence of various diseases. Efficient algorithms exist that perform inference and learning.

Reinforcement learning

Reinforcement learning is concerned with how an agent ought to take actions in an environment so as to maximize some notion of long-term reward. Reinforcement learning algorithms attempt to find a policy that maps states of the world to the actions the agent ought to take in those states. Reinforcement learning differs from the supervised learning problem in that correct input/output pairs are never presented, nor sub-optimal actions explicitly corrected.

Similarity and metric learning

In this problem, the learning machine is given pairs of examples that are considered similar and pairs of less similar objects. It then needs to learn a similarity function (or a distance metric function) that can predict if new objects are similar. It is sometimes used in Recommendation systems.

Genetic algorithms

A genetic algorithm (GA) is a search heuristic that mimics the process of natural selection, and uses methods such as mutation and crossover to generate new genotype in the hope of finding good solutions to a given problem. In machine learning, genetic algorithms found some uses in the 1980s and 1990s. Conversely, machine learning techniques have been used to improve the performance of genetic and evolutionary algorithms.

Rule-based machine learning

Rule-based machine learning is a general term for any machine learning method that identifies, learns, or evolves "rules" to store, manipulate or apply, knowledge. The defining characteristic of a rule-based machine learner is the identification and utilization of a set of relational rules that collectively represent the knowledge captured by the system. This is in contrast to other machine learners that commonly identify a singular model that can be universally applied to any instance in order to make a prediction. Rule-based machine learning approaches include learning classifier systems, association rule learning, and artificial immune systems.

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Feature selection approach

Feature selection is the process of selecting an optimal subset of relevant features for use in model construction. It is assumed the data contains some features that are either redundant or irrelevant, and can thus be removed to reduce calculation cost without incurring much loss of information. Common optimality criteria include accuracy, similarity and information measures.

Pre Lab Exercise:

a. Write 5 applications, that you encounter daily, where Machine Learning is used?

1. Google Search suggestions
2. YouTube/Netflix recommendations
3. Email spam filtering
4. Voice assistants (Google Assistant, Siri)
5. Online shopping recommendations

b. What do you mean by Supervised Learning?

Supervised learning is a type of machine learning where the model is trained using labeled data (input and correct output) to predict results.

c. What do you mean by Unsupervised Learning?

Unsupervised learning is a type of machine learning where no labeled data is provided and the model finds patterns or groups in data.

d. What do you mean by Reinforcement Learning?

Reinforcement learning is a method where an agent learns by interacting with the environment and receiving rewards or punishments.

Colab Link:

https://colab.research.google.com/drive/1bFcz34kLQkDQ1slWoa6vJ_oOnHHGEeS6?usp=sharing

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Program (Code):

Perform the following tasks:

1. Load a dataset in your IDE

```
import pandas as pd
import seaborn as sns

# Load dataset
df = sns.load_dataset('tips')
```

```
# Display first 5 rows
df.head()
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

2. Observe the statistics of all the features

```
# Statistical summary
df.describe()
```

	total_bill	tip	size
count	244.000000	244.000000	244.000000
mean	19.785943	2.998279	2.569672
std	8.902412	1.383638	0.951100
min	3.070000	1.000000	1.000000
25%	13.347500	2.000000	2.000000
50%	17.795000	2.900000	2.000000
75%	24.127500	3.562500	3.000000
max	50.810000	10.000000	6.000000

3. Obtain the shape of the dataset

```
# Shape (rows, columns)
df.shape
```

```
•••  (244, 7)
```

4. Separate all the features

```
# Separate numerical and categorical features

numerical_features = df.select_dtypes(include=['int64', 'float64'])
categorical_features = df.select_dtypes(include=['object', 'category'])
```

```
print("Numerical Features:\n", numerical_features.head())
print("\nCategorical Features:\n", categorical_features.head())
```

```
•••  Numerical Features:
        total_bill   tip  size
    0        16.99  1.01     2
    1        10.34  1.66     3
    2        21.01  3.50     3
    3        23.68  3.31     2
    4        24.59  3.61     4

    Categorical Features:
           sex smoker  day    time
    0  Female     No  Sun  Dinner
    1    Male     No  Sun  Dinner
    2    Male     No  Sun  Dinner
    3    Male     No  Sun  Dinner
    4  Female     No  Sun  Dinner
```

5. Fill the missing values, if any, using the statistically relevant value

```
# Check missing values
df.isnull().sum()

# Fill missing values
# Numerical → median
df[numerical_features.columns] =     df[numerical_features.columns].fillna(df.median(numeric_only=True))
# Categorical → mode
for col in categorical_features.columns:
    df[col].fillna(df[col].mode()[0], inplace=True)
# Verify
df.isnull().sum()
```

```
df[col].fillna(df[col].mode()[0], inplace=True)
```

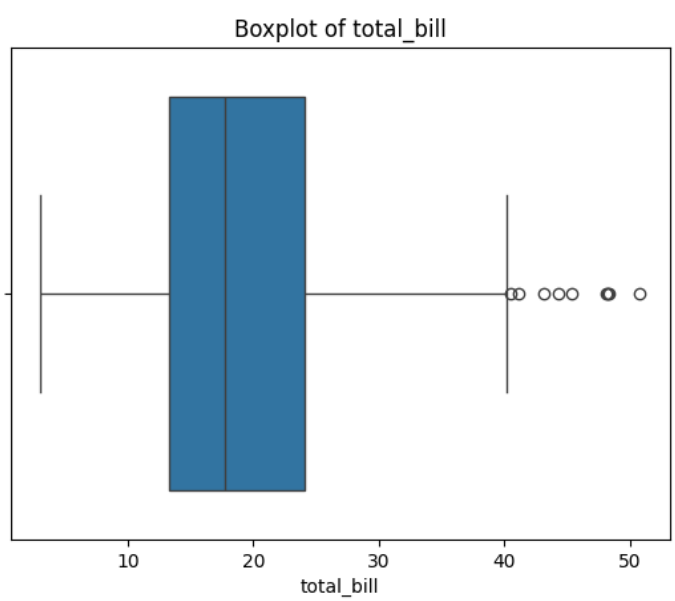
	0
total_bill	0
tip	0
sex	0
smoker	0
day	0
time	0
size	0

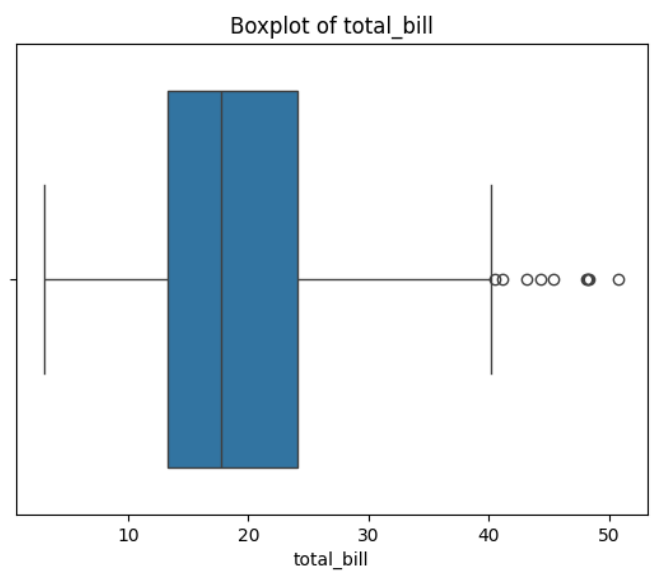
dtype: int64

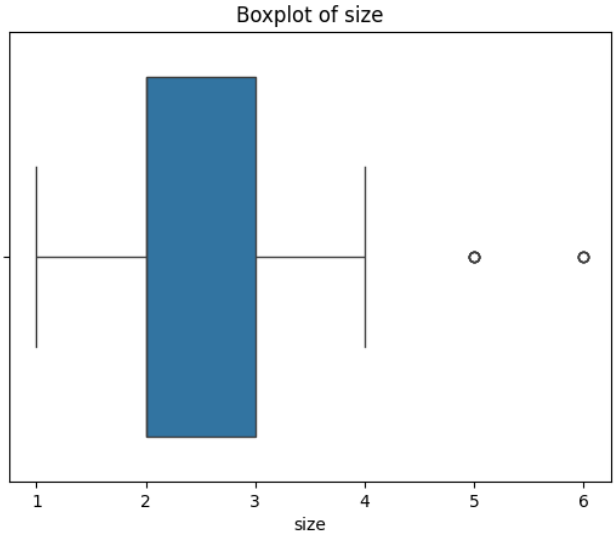
6. Observe the Box-Plot of each feature

```
import matplotlib.pyplot as plt

# Boxplot for all numerical features
for col in numerical_features.columns:
    plt.figure()
    sns.boxplot(x=df[col])
    plt.title(f"Boxplot of {col}")
    plt.show()
```







7. Comment on the IQR range and outliers

```
Q1 = numerical_features.quantile(0.25)
Q3 = numerical_features.quantile(0.75)
IQR = Q3 - Q1

print("IQR:\n", IQR)
```

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Observation and Result Analysis:

a. Statistics of the dataset

The dataset's statistical summary includes measures such as mean, standard deviation, minimum, maximum, and quartiles. These metrics help in analyzing the distribution and variability of the data, and also indicate whether the data follows a normal pattern or is skewed.

b. IQR and Outliers

The Interquartile Range (IQR), defined as Q3 − Q1, captures the spread of the central 50% of the data. Data points that fall below Q1 − 1.5×IQR or above Q3 + 1.5×IQR are treated as outliers, representing extreme values and highlighting variability within the dataset.

Post Lab Exercise:

a. **Which is a better mean of replacing the missing value-mean, median or mode?**

Median is the best method to replace missing values because it is not affected by outliers, while mean is sensitive to extreme values and mode is used for categorical data.

b. **How IQR range helps to understand the nature of the distribution of the data?**

IQR helps to understand the spread of the data and identify outliers, showing whether the data is concentrated or widely distributed.

c. **Which are the ways to replace the categorical data to the numerical ones?**

Categorical data can be converted into numerical form using methods like Label Encoding, One-Hot Encoding, and Ordinal Encoding.